ALLA NIANG | Data Engineer | Analytics Engineering

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SUMMARY

Data Engineer who builds and runs ELT pipelines (Airflow, dbt, SQL, Python) that turn raw, messy source data into tested, trustworthy mart tables for analytics and decision-making. Hands-on with dimensional modeling (star schemas, fact and dimension tables), layered Bronze/Silver/Gold warehouses on Snowflake and DuckDB, data quality testing in dbt, and infrastructure as code with Terraform on AWS (S3, IAM) and Snowflake. Experienced in modeling price and cost data (electricity tariffs, internet offers) into comparable unit indicators and presenting them in dashboards for technical and non-technical audiences. Known by peers for rigorous, well-documented pipelines and clear explanations of technical concepts. Motivated to bring this foundation to FinOps: making cost, usage and margin numbers reconcile and be owned across teams, in a fully remote, async, English-speaking environment.

SKILLS

- **Data engineering & modeling:** dbt (models, tests, seeds), advanced SQL, dimensional modeling (star schema, fact/dimension design), ELT, Bronze/Silver/Gold layering, data quality testing, Apache Airflow, Airbyte, Apache Spark, Apache Kafka
- **Warehouses & databases:** Snowflake, DuckDB, PostgreSQL, MySQL, MongoDB, HDFS
- **Cloud & infrastructure as code:** AWS (S3, IAM), Google Cloud Platform, Terraform, Docker / Docker Compose, MinIO, Git, GitHub Actions (CI/CD)
- **Programming & automation:** Python (pipelines, Scrapy, Playwright, pdfplumber, API integration), SQL
- **Analytics & BI:** Power BI (DAX, Power Query, data modeling), Streamlit, Grafana, Amplitude (product analytics), ELK monitoring
- **AI / ML:** RAG, LLM tool calling, guardrails, non-regression testing (ElevenLabs), PyTorch, OpenCV, FaceNet, SVM

EXPERIENCE

› Data & AI Engineer, Cuberfit, Dakar (Mar 2026 to Present)

- Design and build automated ELT pipelines orchestrated with Apache Airflow to feed the company's analytics platforms.
- Model and transform data with SQL and dbt; deploy workloads in containerized environments with Docker.
- Designed the target ingestion architecture (Airbyte to warehouse to dbt to marts) and prepare the flows for production.
- Run product analytics with Amplitude, working with Product and Engineering teams to turn business needs (including user event tracking) into data deliverables.
- Build CuberOne, Cuberfit's production voice AI assistant (ElevenLabs): RAG over a knowledge base, tool calling into the CRM and customer support (data lookups, ticket creation), guardrails against hallucinated answers, and a non-regression test suite; architecture documented at each step.

› Full-Stack Web Developer (Remote), DSCALE.io, Medad Holding, Dubai (Jun 2024 to Nov 2024)

- Contributed to a blockchain-based (ICP) crowdfunding platform with React and Next.js; integrated data via APIs and built views to track KPIs defined with the team.
- Worked in an international remote team, with all collaboration and deliverables in English.

› AI Training Data Contributor (Freelance, Remote), Appen (Oct 2023 to Oct 2024)

- Produced AI training data (transcription) and performed product testing for remote clients.

› Full-Stack Web Developer, IT Department (DSI), Université Iba Der Thiam, Thiès (Mar 2022 to Oct 2022)

- Built two platforms (claims management, student journey tracking) used by 5,000+ users, cutting processing time by 50% and simplifying reporting for 2,000+ students per semester.

SELECTED PROJECTS

› WattWatch Africa: energy affordability ELT pipeline (AWS S3, Snowflake, dbt, Terraform) (2026, ongoing)

- Bronze/Silver/Gold ELT: Scrapy + pdfplumber ingestion of regulator (CRSE) tariff PDFs for 5 Senegalese electricity operators, raw versioned landing in AWS S3, Airflow COPY INTO Snowflake, dbt transformations tested on every run, Power BI serving.
- Infrastructure as code: S3 bucket, IAM and Snowflake storage integration declared in Terraform, nothing configured by hand in consoles; local Docker/MinIO stack; GitHub Actions CI (lint, Python tests, dbt parse).

• Models tariff data into cost indicators: cost per consumption tier, share of household income, prepaid vs post-paid gap, regional comparison. github.com/niangalla/WattWatch_Africa

› **Observatoire Internet: internet price comparison, dbt + DuckDB star schema (2026, ongoing)**

• Playwright scrapers archive dated, immutable HTML/JSON source snapshots; dbt on DuckDB cleans them into a star schema (dimensions + fact_tarif) where prices live as dated facts, not dimension attributes.

• Normalizes offers into unit-price comparisons (price per Mb/s) across operators and countries; conflicting source pages are kept as separate, traceable records. Power BI and Streamlit reporting.
github.com/niangalla/observatoire-internet

› **DataFlowEngineering360: maritime big data pipeline, Sonatel Academy (2025)**

• Distributed pipeline on Port Autonome de Dakar data with Kafka, Spark and HDFS; Airflow-orchestrated ETL into a PostgreSQL data warehouse; Docker Compose deployment and ELK monitoring.
github.com/niangalla/DataFlowEngineering360_BigData_Maritime_Pipeline

› **DataFlow360 MeteoStream: real-time and batch weather data pipeline (2025)**

• Combined streaming and batch processing with monitoring (ELK) and DevOps practices.
github.com/niangalla/dataflow360_meteo_stream_pipeline

› **Power BI KPI dashboards: resilient agriculture and community health programs, Senegal (2026)**

• Target vs actual tracking (completion rate, gap to target, territorial prioritization) with relational models and DAX measures across 8 regions / 8 districts.
github.com/niangalla/dashboard-agriculture-resiliente-senegal
github.com/niangalla/dashboard-sante-communautaire-senegal

› **Realtime Face Recognition: real-time biometric identification of learners**

• Built a real-time face recognition system with FaceNet embeddings and an SVM classifier (PyTorch, OpenCV).
github.com/niangalla/Realtime_Face_Recognition

EDUCATION

• **Master's in Computer Science, Software Engineering**, Université Iba Der Thiam, Thiès (2023 to present)

• **Data Development Program**, Sonatel Academy, Orange Digital Center, Dakar (Feb 2025 to Dec 2025, graduated)

• **Bachelor's (Licence) in Computer Science, Software Engineering**, Université Iba Der Thiam, Thiès (2018 to 2021): ranked 2nd in class, honors (Mention Bien)

CERTIFICATIONS

• IBM: Data Engineering Essentials
• IBM: NoSQL Databases Essentials
• Retrieve Data with Multiple-Table SQL Queries, Coursera
• EF SET English Certificate (C2)
• Professional Mentorship Program, Global Mentorship Initiative

LANGUAGES

• French: Native · English: Professional (EF SET C2 certified)

VOLUNTEERING

• Volunteer, Marmite du Cœur solidarity campaign (Sonatel Group / Orange Digital Center), Dakar (Mar 2025): food kit distribution during Ramadan.

• Volunteer, 100,000 Students Against COVID-19 program (Ministry of Health), Senegal (Mar to Sep 2020): national awareness campaign.